

Internship Training Program

Day 1 Core Topics, Tasks & Practical Assignments

Welcome to your first day! Below you will find the explicitly structured technical tracks, foundational topics, and hands-on assignments required for Day 1. Please locate your respective role domain and complete all objectives before the end of the day.

ENGINEERING TRACK

1. Java Full Stack Interns – Day 1 Topics & Assignments

Core Focus Areas & Topics:

- **Environment & Toolchain Setup:** Installation and configuration of JDK 17+, Apache Maven, and Integrated Development Environments (IntelliJ IDEA or Eclipse).
- **Java Language Fundamentals:** Deep dive into strongly-typed primitives, wrapper classes, core arithmetic and bitwise operators, and structural control flow (`if-else`, `switch-case`, loops).
- **Object-Oriented Architecture (Foundations):** Classes, object instantiation, memory allocation (Stack vs. Heap), variables scoping, and custom constructor design.
- **Web UI Basics:** Standard semantic HTML5 layout structuring and CSS3 selector mechanisms, specific to full-stack front-end interfaces.

PRACTICAL ASSIGNMENT TASKS

Task 1.1 (Backend): Write a comprehensive Java console application that takes user inputs to evaluate parameters and execute dynamic calculations (e.g., a multi-tier tax calculator or an inventory tracking logic). Leverage explicit class definitions and parameterized constructors.

Task 1.2 (Frontend): Develop a clean, static structural landing page layout for an e-commerce dashboard platform using clean HTML5 tags and descriptive CSS grid/block properties.

2. MERN Stack Interns – Day 1 Topics & Assignments

Core Focus Areas & Topics:

- **Modern Ecosystem Configuration:** Installing Node.js LTS, configuring Node Package Manager (npm), and environment setup for Visual Studio Code.
- **Advanced ECMAScript 6+ Fundamentals:** Core mastery over modern JS constructs including `let/const`, arrow syntax, template literals, array destructuring, and Rest/Spread operators.
- **Asynchronous JS Foundation:** Understanding Single-Threaded Event Loop execution, Callbacks, and basic Promise flows.
- **Front-End Architecture:** Semantic layouts, CSS custom variables, and fundamental structural frameworks.

PRACTICAL ASSIGNMENT TASKS

Task 2.1 (Algorithmic Scripting): Build a localized JavaScript script implementing data parsing manipulation algorithms (e.g., processing a mock raw JSON string array of user objects to extract, filter, and clean specific records based on target criteria).

Task 2.2 (Component UI): Create a responsive user profile profile card structure with explicit vanilla layout rules, ensuring adaptive fluid scaling across desktop and mobile form factors.

3. Data Analyst Interns – Day 1 Topics & Assignments

Core Focus Areas & Topics:

- **Analytical Framework & Tooling:** Workspace configuration inside Microsoft Power BI Desktop and relational execution in structured environments (SQL Server / PostgreSQL).
- **Structured Query Language (DDL/DML):** Writing robust database extraction scripts using standard `SELECT`, `WHERE` criteria filters, and baseline relational joins (`INNER`, `LEFT`, `RIGHT`).
- **Power BI Data Ingestion:** Importing flat transactional files (.csv, .xlsx), analyzing data shapes, and identifying explicit data types.
- **Foundational Data Cleaning:** Identifying missing dimensions, duplicates, and introducing column transformations.

PRACTICAL ASSIGNMENT TASKS

Task 3.1 (SQL Querying): Construct a functional SQL script file containing queries that extract key subsets from a provided relational sales schema (e.g., retrieving high-value orders, calculating active dates, and joining user info profiles).

Task 3.2 (Power BI Mashup): Load a raw multi-tab spreadsheet dataset into Power BI Desktop, perform basic column formatting/renaming optimizations, and organize a preliminary summary dashboard with 3 distinct exploratory charts.

4. Business Analyst Interns – Day 1 Topics & Assignments

Core Focus Areas & Topics:

- **Business Analysis Foundations:** Introduction to the SDLC framework, business process lifecycle modeling, and the specific role of a BA in cross-functional squads.
- **Requirement Engineering Concepts:** Dissecting, defining, and sorting Functional Requirements vs. Non-Functional Requirements (FRs vs. NFRs).
- **Analytical Documentation:** Overview of Business Requirement Documents (BRD) and Functional Specification Documents (FSD).
- **Advanced Spreadsheet Modeling:** Structuring logical operational tables, conditional lookups (`VLOOKUP`, `XLOOKUP`), and summary aggregations via Pivot Tables.

PRACTICAL ASSIGNMENT TASKS

Task 4.1 (Requirement Documenting): Review a case narrative for a standard online booking application. Draft a brief functional matrix clearly itemizing 5 essential Functional Requirements and 3 corresponding Non-Functional Requirements.

Task 4.2 (Spreadsheet Synthesis): Analyze a raw corporate operations dataset using spreadsheet formulas, build a structured Pivot Table summarizing key product lines, and highlight performance margins.

End-of-Day Deliverables

All completed work must be formally packaged, compiled, and submitted by **6:00 PM** today following the guidelines below:

- **Code Repositories (Java / MERN):** Commit your finalized project code structure to a localized repository or clean workspace directory, properly documenting instructions in a `README.md`.
- **Analytics & Scripts (Data Analysts):** Package your structured `.sql` query script file along with your compiled `.pbix` Power BI workspace dashboard.
- **Documentation & Models (Business Analysts):** Export your formatted Requirements Matrix document as a PDF, accompanied by your `.xlsx` analytical spreadsheet model.
- **Submission Portal:** Upload your deliverables to the designated team portal link shared by your supervisor, ensuring files are clearly named using the syntax:
[YourName]_[TrackName]_Day1_Submission.